

Aayam Regmi

Sallaghari, Bhaktapur, Nepal | +977 9805391127 | aayam.regmi2003@gmail.com
linkedin.com/in/aayam-regmi-196193283 | github.com/AayamRegmi | aayamregmi.com.np

RESEARCH INTERESTS

Computer vision, robotic perception, and vision-based human-computer interaction, with a focus on pose estimation, real-time sensor fusion, and deep learning systems applied to physical-world environments. Seeking to pursue graduate research at the intersection of robotics and computer vision.

EDUCATION

B.Sc. (Hons) Computer Science, Database Specialization

Oct 2022 – May 2026 (Results Pending)

Leeds Beckett University, via The British College, Kathmandu, Nepal (UK-accredited)

- GPA: ~3.9 / 4.0 est. (82–91% across all semesters; final results pending)
- Top 5% of graduating cohort
- 91% in 7th Semester; Merit / Refund Scholarship candidate
- Selected for International Student Exchange Programme (1 of ~200 students)

A Levels

Sep 2020 – May 2022

Trinity International College, Kathmandu, Nepal

Secondary Education Examination (SEE)

Apr 2012 – Apr 2020

Galaxy Public School, Bhaktapur, Nepal

RESEARCH

Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction

2026

Lead Researcher, The British College / Leeds Beckett University

- Benchmarked MediaPipe and OpenPose (Convolutional Pose Machines) for real-time piano key-press detection via pose estimation and geometric containment.
- Designed and built a custom MIDI-video synchronisation pipeline; collected a dataset of 38 sessions comprising 4,777 annotated note-on events with Fitzpatrick skin-type, hand-size, and illuminance metadata.
- Proposed horizontal-axis Mean Per Joint Position Error (MPJPE) as a domain-specific localization metric to evaluate lateral fingertip precision.
- Achieved 63.3% polygon hit rate with MediaPipe vs 8.1% with OpenPose; identified a left-right hand detection asymmetry (95.8% right vs 53.2% left) informing future dataset balance requirements.
- Published the de-identified dataset (38 sessions: MIDI note events, per-key calibration, and pose-accuracy results for MediaPipe/OpenPose) as an open benchmark on Zenodo under CC BY 4.0: zenodo.org/records/21201358
- Manuscript in preparation for conference submission.

PROJECTS

Tree Conservation Monitoring System

2025

International Student Exchange Programme, 1st Place (Winning Team)

- Designed and assembled a custom battery-powered IoT device around an Arduino Mega ADK integrating a NEO-6M GPS module, DollaTek UART 2D barcode scanner, I2C SSD1306 OLED display, MicroSD card reader, and a 4-key input board over UART, I2C, and SPI buses.
- Device captures GPS coordinates, timestamps, and field metrics per tree; logs records to MicroSD for offline resilience and syncs to a central database for persistent tree-level history.
- Deployed physical barcodes on trees in the field; scanning a barcode with the handheld device instantly retrieves that tree's full record on the OLED display, with no network required at scan time.
- Technologies: Arduino (C++), NEO-6M GPS, UART/I2C/SPI protocols, MicroSD storage, OLED display, barcode scanning, relational database.

Credit Card Default Prediction

2024

Machine Learning Project

- Built and compared classification models (Random Forest, Logistic Regression, SVM, Gradient Boosting) on the UCI credit card default dataset; achieved 92% accuracy with Random Forest.
- Applied feature engineering, SMOTE oversampling for class imbalance, and cross-validated hyperparameter tuning.
- Technologies: Python, scikit-learn, Pandas, NumPy, Matplotlib.

Open Source: ats-resume-checker, a deterministic ATS resume-screening simulator (Python CLI + in-browser app); a full e-commerce website with authentication and backend; a crime-tracking database system on Oracle; a banking system; and more at github.com/AayamRegmi.

WORK EXPERIENCE

Full-Stack Software Developer Intern, Professional Computer Systems, Kathmandu, Nepal

Nov 2024 – Mar 2025

- Developed RESTful backend services and authentication flows using Spring Boot and Spring Security with a PostgreSQL database.
- Built responsive frontend components in ReactJS; collaborated on production deployment pipeline.

TECHNICAL SKILLS

Computer Vision & ML: PyTorch, TensorFlow, OpenCV, MediaPipe, OpenPose, scikit-learn, Convolutional Neural Networks (CNN), pose estimation

Data Science: NumPy, Pandas, Matplotlib, SciPy, Jupyter

Programming Languages: Python (advanced), Java, JavaScript, SQL, PHP, Lua

Web & Backend: Spring Boot, Spring Security, React, Node.js, Django, REST APIs

Databases: PostgreSQL, Oracle, PL/SQL

Embedded & Hardware: Arduino (C++), NEO-6M GPS, UART / I2C / SPI, OLED displays, MicroSD, barcode scanners

Tools & Environments: Git, Linux/Ubuntu, Google Colab, Jupyter

AWARDS & HONORS

- 1st Place, International Student Exchange Programme Project Competition (2025)
- International Student Exchange Programme Selectee, chosen from 200+ applicants (2025)
- Top 5% of Graduating Cohort, Leeds Beckett University / The British College (2026)
- Merit / Refund Scholarship Candidate, 91% score, 7th Semester (2026)

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** (Coursera / Stanford & DeepLearning.AI, 2023)
- **Advanced Learning Algorithms** (Coursera / Stanford & DeepLearning.AI, 2023)
- **Unsupervised Learning, Recommenders, Reinforcement Learning** (Coursera / Stanford & DeepLearning.AI, 2023)
- **Django Web Development** (Mindrisers Institute of Technology, 2023)
- **MERN Stack Development** (The British College, 2023)